

Solution Architecture

Date	29/09/2026
Team ID	-
Project Name	BlogNest - AI-Powered Blog Management Backend
Maximum Marks	-

Solution architecture bridges the gap between the business problem and the technical solution. BlogNest uses a layered server architecture: every request is filtered by authentication and role checks before it reaches the business logic and data layer.

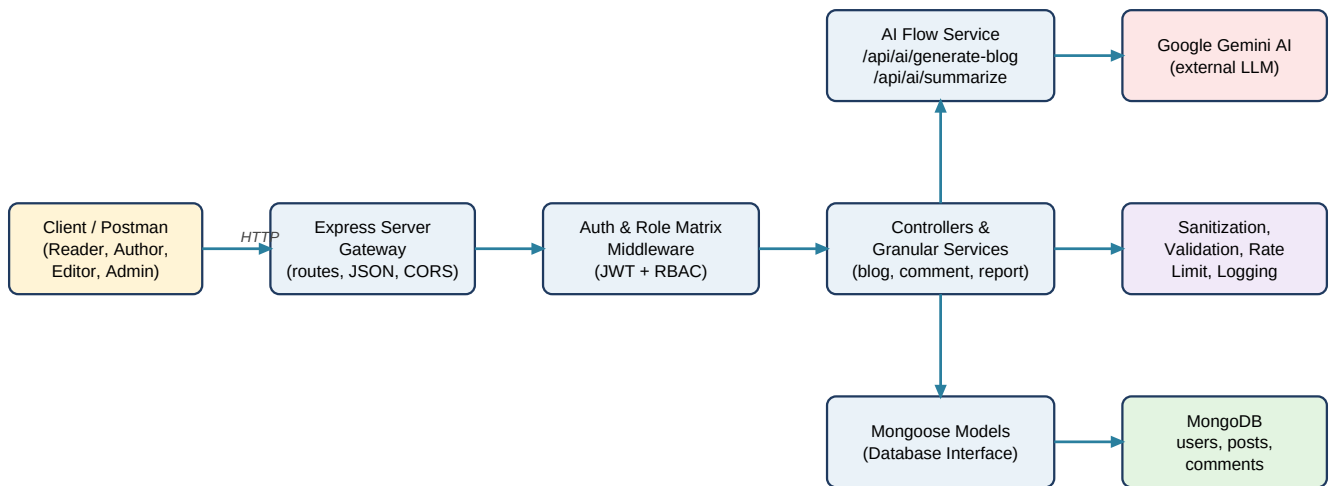


Figure 1: BlogNest architecture and data flow

Components:

- **Interface:** Postman or any frontend sends HTTP requests (GET, POST, PUT, DELETE) and receives JSON.
- **Express Server Gateway:** listens on the port, maps routes, parses JSON payloads and handles CORS.
- **Authentication & Role Matrix Middleware:** extracts the token, validates its signature and checks that the role is allowed for the route.
- **Granular Services (Controllers):** enforce blog lifecycle rules, comment queues and reporting.
- **Database Interface Layer:** Mongoose schemas and models translate operations into MongoDB queries.
- **AI Flow:** connects to Google Gemini via /api/ai/generate-blog and /api/ai/summarize to offload blog generation and summarization.
- **Data model:** User to Post, User to Comment, and Post to Comment are all one-to-many relationships.